Liquidity and Price Impact in Two Uniswap Pools

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1 The Core Idea

Market liquidity is the cost of trading a given size. On an automated market maker (AMM) the cost is set by the pool reserves, so it can be read from the chain. The project compares two pools that sit at opposite ends of the scale: a new, almost empty pool (HLD/ETH, Uniswap V4 on Base) and a deep one (ETH/USDC, Uniswap V3 on Ethereum; Adams et al., 2021). A risk manager cares about the same question for both: what does it cost to get out?

The two pools need two different treatments:

- **HLD/ETH is modelled:** it has almost no trades, so returns and volatility cannot be estimated. Its costs are computed from the real reserves with the constant-product formula.
- **ETH/USDC is estimated:** it has thousands of trades, so price impact is measured from real swap events.

2 Why It Matters

- **Size relative to reserves sets the cost.** A 1 EUR order in HLD/ETH takes 16.7% of the HLD reserve and pays 2,126 bp. Only 101 bp of that is the 1% pool fee. The rest is the curve.
- **Depth gaps are enormous.** Moving the price by 1% takes 1.5×10^{-5} ETH in the small pool and 1,618 ETH in the deep one, a factor of about 10^8. Volatility, the usual risk input, says nothing about this difference.
- **AMM impact is mechanical.** The price is a function of the reserves, so the cost of a trade is known before it is sent. That helps a risk manager, but it also means a high R^2 in an impact regression measures the pool's depth, not the presence of informed traders.
- **A risk number depends on how it is sampled.** Realised volatility of the ETH/USDC pool is 24.2% annualised with the repository's per-swap scaling and about 40% on 5-minute bars. Each is correct for its own interval, and only one fits a 5-minute risk limit.

3 Assumptions

Every item below is built into the code. The last sentence of each says what moves if it is wrong.

- **HLD/ETH snapshot.** One read of the pool state at Base block 49,023,454 (2026-07-23 19:50:55 UTC): $\sqrt{P}$, tick, LP fee (10,000 pips, 1%), protocol fee 0, liquidity L. No hook is

attached (zero address). A snapshot misses later deposits or withdrawals, and the depth of such a small pool can change by an order of magnitude with one liquidity provider.

- **Constant product for V4.** The model treats the active liquidity L as constant over the whole trade, with virtual reserves $x = L/\sqrt{P}$ and $y = L\sqrt{P}$. The pool has a tick spacing of 200, about 2.02% of price per spacing, and a 1 EUR order moves the price by 44%, across about 18 spacings. Real liquidity outside the current range is unknown, so the true cost could be lower or much higher.

- **Prices in EUR.** The HLD price comes only from the pool, and ETH is valued at 1,645.88 EUR (CoinGecko, 2026-07-23). One HLD is worth 7.48×10^{-8} EUR. A different ETH price rescales the order size in ETH, so a fixed EUR order changes its share of the reserve.

- **ETH/USDC sample.** 6,737 swaps of the 0.05% pool on Ethereum between blocks 25,587,478 and 25,598,266 (22 July 09:48 UTC to 23 July 21:58 UTC, 36 hours), prices from $\sqrt{P}$ after each swap, converted to dollars with an ETH price of \$1,881.94 (CoinGecko). The conversion enters λ squared, so a 2% error in the ETH price moves the dollar λ by 4%.

- **Linear impact.** Kyle's (1985) regression is a straight line in signed flow, fitted on consecutive swaps with ordinary standard errors. Swaps in the same block are separate observations. Errors are probably heteroskedastic and correlated, so the t-statistic is too large.

- **Time scaling.** Volatility is the per-swap standard deviation scaled by the mean gap between swaps with a positive time difference (28.5 s). One third of the gaps are zero (same block), and the mean over all gaps is 19.3 s. A shorter gap would raise the annualised figure.

- **Mid price.** The effective spread takes the price after the previous swap as the mid. A swap in the same block as the one before it uses that swap's price. The measure mixes the fee, the swap's own impact and any move between swaps.

- **Linear extrapolation.** The cost of a 1% move is $1\% \times P/\lambda$. This assumes impact stays linear well outside the observed trades. The largest swap in the sample is 430 ETH, and the 1% move needs 1,618 ETH.

- **Single regime.** Thirty-six hours of one pool in July 2026. Another regime, with higher volatility or thinner liquidity, would give a larger λ.

4 Method

4.1 Reading the V4 Pool State

A Uniswap V4 pool is identified by the keccak256 hash of its key (two currencies, fee, tick spacing, hooks). The pool state sits in a mapping of the `PoolManager` contract, so the code computes the storage slot and reads it with `extsload`. The first word packs four fields: the square-root price $\sqrt{P} \cdot 2^{96}$ in the low 160 bits, then the tick (24 bits), the protocol fee (24 bits) and the LP fee (24 bits). The code decodes them with bit masks. The liquidity L is read from a second slot. The tick and the square-root price agree to within 0.01%, which checks the decoding.

4.2 Constant-Product Model

From L and $\sqrt{P}$ the virtual reserves of the pool are

$$x = \frac{L}{\sqrt{P}}, \qquad y = L\sqrt{P}, \qquad k = xy, \tag{1}$$

with x in ETH and y in HLD. In this pool $x = 0.003$ ETH and $y = 65{,}992{,}981$ HLD. For an order of Δx ETH and a fee $\phi = 1\%$, the code (`src/amm.py`) computes

$$x' = x + \Delta x(1-\phi), \qquad y' = \frac{k}{x'}, \qquad \Delta y = y - y', \tag{2}$$

$$\text{slippage} = \frac{\Delta x/\Delta y}{x/y} - 1. \tag{3}$$

The ETH needed to push the price up by a fraction p is $x(\sqrt{1+p}-1)/(1-\phi)$. The impermanent loss of a 50/50 position, for a price ratio r, is $2\sqrt{r}/(1+r)-1$ (-5.72% at $r = 2$, -20.0% at $r = 4$). Slippage rises with the order's share of the reserve, and it never falls below $1/(1-\phi) - 1 = 101$ bp. Pool depth is the only parameter: doubling both reserves halves the share of any given order.

4.3 Estimators on Real Swaps

For the ETH/USDC swaps (`src/microstructure.py`), with q_i the signed ETH flow of swap i (positive when the taker buys ETH):

- **Kyle's lambda:** the OLS slope of the price change on the signed flow, $\Delta p_i = \alpha + \lambda q_i + \varepsilon_i$, between consecutive swaps. A binned version sums the flow and takes the price change over windows of 60 and 300 seconds.
- **Realised volatility:** the standard deviation of log price changes per swap, scaled to a year with the average time between swaps $\bar{dt}$:

$$\hat{\sigma} = s_{\Delta \ln p}\sqrt{\frac{365 \cdot 24 \cdot 3600}{\bar{dt}}}. \tag{4}$$

- **Effective spread:** for each swap, twice the gap between the execution price and the mid price before the swap, in basis points:

$$\text{ES}_i = 2\,\frac{|P_i^{\text{exec}} - P_{i-1}^{\text{mid}}|}{P_{i-1}^{\text{mid}}} \times 10^4. \tag{5}$$

The pool quotes WETH per USDC, so the fitted λ is converted to dollars per ETH with $\lambda_\$ = |\lambda|\,P^2$, where P is the ETH price in dollars (since $P = 1/p$ and $dP = -P^2 dp$).

4.4 Cross-Check from Pool Liquidity (comparison, not in the code)

A V3 pool has the same constant-product curve inside its active range, so the regression can be tested against the pool's own liquidity. With L and $\sqrt{P}$ from each swap, the virtual WETH reserve is $Y = L\sqrt{P}/10^{18}$, the ETH needed for a $+1\%$ move is $Y(1-1/\sqrt{1.01})$, and $\lambda_\$ \approx 2P/Y$.

5 Method Comparison

Method	Strength	Weakness
Constant-product model	Needs one snapshot, works with no trades	Assumes constant liquidity across the move
Kyle's lambda per swap	Uses every swap, high precision	Mechanical in an AMM; noisy at the swap level
Kyle's lambda in bins	Less sensitive to swap-level noise	Fewer points; depends on the bin length
Effective spread	Direct cost of a trade, in bp	Mixes fee, impact and drift between swaps

The model is the only option when a pool has no trades, and it describes the curve, not what trades will actually pay. With enough swaps, the estimators measure what trades actually paid. The spread suits small trades, and λ suits large ones, because it extends the cost to sizes the sample did not contain.

6 Results

HLD/ETH uses the pool state at Base block 49,023,454. ETH/USDC uses 6,737 swaps over 36 hours (22–23 July 2026), with an average of 28.5 seconds between swaps (positive gaps).

Metric	HLD/ETH (modelled)	ETH/USDC (estimated)
Trades observed	about 0	6,737 in 36 h
Cost of a small order	1 EUR order: 2,126 bp	10.2 bp (median effective spread)
Capital to move the price +1%	1.5×10^{-5} ETH (about 0.025 EUR)	1,618 ETH (about $3.05M)
Kyle's lambda	not estimable	0.0116 $/ETH per ETH
t-statistic, R^2 of the fit	not applicable	317, 0.94
Realised volatility	not identifiable	24.2% annualised

- **HLD/ETH:** the pool value is about 10 EUR (twice the 0.003 ETH reserve at 1,645.88 EUR per ETH). The LP fee is 1%.
- **ETH/USDC:** the mean effective spread is 15.7 bp, above the median of 10.2 bp, so the distribution is skewed to the right.
- **Depth gap:** the capital needed to move the price by 1% differs by a factor of about 10^8 between the pools.

The median effective spread of 10.2 bp is almost exactly twice the 5 bp fee of the pool. A median swap pays the fee and little else: it is small against the depth, and its impact is a fraction of a basis point. Size matters in the tail. Recomputed from the CSV, the median spread is 10.0 bp for the smaller half of swaps, 10.8 bp from the 50th to the 90th percentile, 14.3 bp from the 90th to the 99th and 22.8 bp for the largest 1% (68 swaps, median size 208 ETH). The mean of 15.7 bp is also pulled up by tiny swaps (median 0.07 ETH) whose execution price is noisy: their mean is 19.4 bp against a median of 10.0. The two halves of the sample agree on the median (10.16 and 10.18 bp) but not on the mean (14.2 and 17.2 bp), so the tail is the unstable part.

How the HLD/ETH cost grows with order size (EUR order, ETH at 1,645.88 EUR):

Order (EUR)	0.1	1	5	10
Slippage (bp)	304	2,126	10,227	20,354
Share of HLD reserve (%)	2.0	16.7	50.1	66.7

The cost is roughly linear in size for small orders and bends only because the reserve is finite. A 10 EUR order, equal to the whole pool value, pays 203% over the spot price and still takes only two thirds of the HLD. This is a statement about the formula. It does not say that anyone could sell 10 EUR of HLD back at a sensible price.

The ETH/USDC depth estimate depends on the estimator. The price of ETH used for conversion is $1,881.94:

Estimator	$\lambda_{\$}$	R^2	Observations	ETH for +1%
Per swap	0.0116	0.94	6,736	1,618
60-second bins	0.0075	0.73	1,912	2,495
300-second bins	0.0108	0.92	434	1,739

The estimates range from 1,618 to 2,495 ETH. The 60-second bins are the outlier. A bin's price change starts from the price after its first swap, while its flow includes that swap, so part of the impact is dropped, and the loss is larger when a bin holds few swaps. The cross-check supports the per-swap figure. The pool's own liquidity gives a median of 1,636 ETH for a 1% move (10th to 90th percentile: 741 to 1,755 ETH), about 1% above the regression. The active liquidity changed in 11.7% of the swaps, so the cost of a large order depends on which liquidity is in range when it arrives.

Realised volatility depends on the sampling interval. The repository's figure uses per-swap returns. On fixed bars of the pool price, recomputed from `data/eth_swaps.csv` (not in the code):

Sampling	Per swap	1 minute	5 minutes	1 hour
Annualised volatility (%)	24.2	33.7	40.3	40.6

The per-swap figure is the lowest. Consecutive per-swap returns are positively correlated (0.09 at lag 1 and 0.11 at lag 2), so variance grows faster than the number of swaps and a per-swap scaling understates the longer-horizon figure. Scaling by the mean gap over all swaps (19.3 s) instead of positive gaps only gives 29.4%. For a risk measure over minutes to hours the 5-minute and 1-hour figures are the relevant ones, and they agree at about 40%.

7 Worked Example

A 1 EUR order in HLD/ETH (all amounts in EUR). The reserves are $x = 0.003$ ETH and $y = 65{,}992{,}981$ HLD, so $k = 197{,}978.94$. One HLD is worth 7.482×10^{-8} EUR.

Step 1. $\Delta x = 1/1645.88 = 6.0758 \times 10^{-4}$ ETH.

Step 2. After the 1% fee, $0.99 \cdot 6.0758 \times 10^{-4} = 6.0150 \times 10^{-4}$ ETH enters the curve, so $x' = 0.0036015$ and $y' = 197{,}978.94/0.0036015 = 54{,}971{,}217$.

Step 3. The buyer receives $\Delta y = 65{,}992{,}981 - 54{,}971{,}217 = 11{,}021{,}764$ HLD.

Step 4. The execution price is $1/11{,}021{,}764 = 9.073 \times 10^{-8}$ EUR per HLD against a spot of 7.482×10^{-8}. The slippage is +21.26%, or 2,126 bp. The order takes 16.7% of the HLD reserve,

which explains the size of the cost. After the trade the pool price is 44.1% higher, at 1.078×10^{-7} EUR per HLD.

Moving the price by 1%. $x(\sqrt{1.01} - 1)/0.99 = 1.511 \times 10^{-5}$ ETH, which is about 0.025 EUR.

Sensitivity to depth. If the pool held 10, 100 or 1,000 times the reserves, the same 1 EUR order would cost 304, 121 and 103 bp, and the capital to move the price by 1% would be 0.25, 2.49 and 24.9 EUR. The cost converges to the 101 bp fee, and the 1% move scales one for one with depth. A pool of 100 times the reserves is still worth only about 1,000 EUR.

The same move in ETH/USDC (all amounts in dollars). With $|\lambda| = 3.2834 \times 10^{-9}$ WETH per USDC per ETH, $\lambda_\$ = 3.2834 \times 10^{-9} \cdot 1881.94^2 = 0.011629$ \$/ETH per ETH. A 1% move is \$18.82, which needs 18.82/0.011629 = 1,618 ETH, or \$3.05M. A trade of 10 ETH moves the price by about \$0.116, or 0.62 bp. The largest swap in the sample, 430 ETH, moves it by about \$5.0 (0.27%), so the 1% figure is 3.8 times the largest observed trade.

8 Limits and Extensions

- **Local approximation:** constant product is only a local description of V4 concentrated liquidity. Far from the current price the real curve differs, and a 1 EUR order leaves the range around the current tick.
- **Modelled costs:** the HLD/ETH figures come from a formula and one block of data, not from observed trades. Reserves are virtual, not contract balances.
- **Linear extrapolation:** the \$3.05M figure assumes the impact stays linear for a move far larger than any observed trade. It gives an order of magnitude only.
- **Short sample:** 36 hours of ETH/USDC data say nothing about other market regimes.
- **Measurement:** the effective spread uses the price before the swap as mid, so it mixes the pool fee, the price impact of the swap itself and any price change between swaps.

Three extensions would fix the largest limits. First, read the tick bitmap and the liquidity of each initialised tick, so that the cost of an order is computed across the real liquidity profile and not a constant L. Second, extract the swaps by block and use block-level flows and prices, which removes the zero-gap problem in the time scaling. Third, extend the ETH/USDC sample to several weeks and report λ and the 1% depth for calm and stressed days separately.

9 Conclusion

A pool's price impact can be read from its reserves, and the two pools show how far apart that reading can be: a cost of a few basis points for a median swap in one, and 2,126 bp for a 1 EUR order in the other. For anyone holding a thin token, the exit cost is set by the liquidity profile and not by volatility, and it should be sized before the position is.

References

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